Technical Specification: `BUSTED --auto` and the Saturation Error Sink (`--error-sink`) in HyPhy 3

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Abstract

The Branch-Site Unrestricted Statistical Test for Episodic Diversification (`BUSTED`) tests whether a gene has experienced positive selection on at least one site and at least one branch. Over time, `BUSTED` has accumulated several optional model extensions: synonymous rate variation (`--srv`, $M \in \{1, 2, 3\}$), multi-nucleotide substitutions (`--multiple-hits`, δ and ψ), an alignment error sink (`--error-sink`), and a variable number of non-synonymous rate classes ($K \in \{1, 2, 3\}$). Running the full Cartesian grid of $2 \times 3 \times 3 \times 2 = 36$ model pairs is computationally wasteful and statistically poorly calibrated. Here we specify two additions to HyPhy 3. First, we introduce a parameter-free Huber ϵ-contaminated saturation replacement kernel (`--error-sink`) that quarantines alignment and sequencing artifacts in $O(S)$ time per branch-site. Second, we define a four-gate automatic model selector (`--auto`) that chooses the active model components from a single baseline fit using branch-site surprisal, error-quarantined tangent scores, Neyman-Scott synonymous overdispersion, and constrained-null AIC_c pruning.

1 Problem Statement

A general `BUSTED` analysis includes four discrete model configuration choices:

1. **Error sink** ($E \in \{\text{Off}, \text{On}\}$): whether to include a mixture component that absorbs non-biological alignment or sequencing errors.
2. **Multi-nucleotide hits** ($H \in \{\text{None}, \text{Double}, \text{Double+Triple}\}$): whether to allow instantaneous 2-nucleotide (δ) and 3-nucleotide (ψ) codon substitutions.
3. **Synonymous rate variation** ($M \in \{1, 2, 3\}$): the number of discrete site-to-site synonymous rate categories α_m.
4. **Non-synonymous rate classes** ($K \in \{2, 3\}$): the number of discrete branch-site ω_k categories.

Two statistical failure modes occur when these components are enabled manually or selected by naive AIC_c grid search:

- **Confounding between alignment errors and multi-nucleotide hits:** A local misalignment or frameshift introduces clusters of 2-nt and 3-nt codon differences on a single short branch ($t_b \ll 0.05$). Under standard `BUSTED`, these changes force $\hat{\omega}_K \gg 100$ and cause false-positive selection calls. Under `BUSTED-MH`, the same artifact inflates $\hat{\delta}$ and $\hat{\psi}$ across the entire tree.
- **Boundary competition between ω_K and the error sink:** On a short branch of length $t_b \approx 10^{-3}$, an unconstrained rate class with $\omega_K = 2000$ yields $\exp(\boldsymbol{Q}(\omega_K)t_b) \approx \mathbf{1}\boldsymbol{\pi}^T$, which is numerically indistinguishable from a saturation replacement kernel.

2 Option A: The Huber ϵ-Contaminated Saturation Error Sink

2.1 Formulation of the Replacement Kernel

In HyPhy 2.5, `--error-sink` added a fourth rate class with fixed $\omega_{\text{sink}} = 100$ and weight $p_{\text{sink}} \leq 0.01$. That formulation had three defects. First, its transition probabilities depended on branch length t_b, so it failed to absorb errors on terminal branches of length $t_b \to 0$. Second, it only increased single-nucleotide non-synonymous rates and could not absorb 2-nt or 3-nt frameshift codons in one step unless $\delta, \psi > 0$. Third, it required an extra 61×61 matrix exponential per branch.

In HyPhy 3, we replace the $\omega = 100$ rate class with an exact branch-length-independent off-diagonal replacement kernel $\boldsymbol{P}_{\text{sink}} \in \mathbb{R}^{S \times S}$ ($S = 61$ sense codons):

$$P_{\text{sink}}(i,j) = \begin{cases} 0 & \text{if } i = j, \\ \dfrac{\pi_j}{1 - \pi_i} & \text{if } i \neq j, \end{cases} \tag{1}$$

where $\boldsymbol{\pi} = (\pi_1, \ldots, \pi_S)$ is the empirical $F3 \times 4$ codon equilibrium distribution ($\sum_{j=1}^{S} \pi_j = 1$). Every row of $\boldsymbol{P}_{\text{sink}}$ is non-negative and sums to 1. Under $\boldsymbol{P}_{\text{sink}}$, a corrupted codon i is replaced by a distinct codon $j \neq i$ drawn from the background codon distribution $\boldsymbol{\pi}$, independent of the branch length t_b and independent of the Hamming distance $d(i,j) \in \{1,2,3\}$.

On any tested branch b of length t_b, the effective transition probability matrix for synonymous rate class $m \in \{1, \ldots, M\}$ and non-synonymous rate class $k \in \{1, \ldots, K\}$ is the Huber ϵ-contaminated mixture:

$$\tilde{\boldsymbol{P}}_b(m,k) = (1 - \epsilon_{\text{sink}}) \boldsymbol{P}_{\text{bio}}(t_b \mid \alpha_m, \omega_k) + \epsilon_{\text{sink}} \boldsymbol{P}_{\text{sink}}. \tag{2}$$

Here $\boldsymbol{P}_{\text{bio}}(t_b \mid \alpha_m, \omega_k) = \exp(\alpha_m t_b \boldsymbol{Q}(1, \omega_k))$ is the biological transition matrix, and $\epsilon_{\text{sink}} \in [0, \epsilon_{\max}]$ ($\epsilon_{\max} = 0.05$) is the alignment-wide error proportion shared by the Unconstrained Alternative and Constrained Null models.

2.2 Zero-Matrix-Exponential $O(S)$ Bilinear Evaluation

Because $\boldsymbol{P}_{\text{sink}}$ is a rank-1 matrix minus its diagonal, evaluating the branch-site conditional likelihood under $\boldsymbol{P}_{\text{sink}}$ requires no matrix exponential and no $O(S^2)$ matrix-vector multiplication. Given the post-order child conditional likelihood vector $\boldsymbol{d} = D_{v,p} \in \mathbb{R}^S$ and the pre-order outside (ancestral complement) vector $\boldsymbol{u} = V_{v,p} \in \mathbb{R}^S$ at branch b and pattern p, first compute the scalar dot product:

$$S_\pi(\boldsymbol{d}) = \sum_{j=1}^{S} \pi_j d_j = \boldsymbol{\pi}^T \boldsymbol{d}. \tag{3}$$

The action of $\boldsymbol{P}_{\text{sink}}$ on $\boldsymbol{d}$ at ancestral state i is:

$$(\boldsymbol{P}_{\text{sink}} \boldsymbol{d})_i = \sum_{j \neq i} \frac{\pi_j d_j}{1 - \pi_i} = \frac{S_\pi(\boldsymbol{d}) - \pi_i d_i}{1 - \pi_i}, \tag{4}$$

and the complete bilinear form across branch b is evaluated in $O(S)$ operations:

$$L_{b,p}^{\text{sink}} = \boldsymbol{u}^T \boldsymbol{P}_{\text{sink}} \boldsymbol{d} = \sum_{i=1}^{S} u_i \frac{S_\pi(\boldsymbol{d}) - \pi_i d_i}{1 - \pi_i}. \tag{5}$$

2.3 EM Responsibility Update and ω_K Upper-Bound Separation

Let $L^{\text{bio}}_{b,p} = \boldsymbol{u}^T \left(\sum_{m,k} q_m p_k \boldsymbol{P}_{\text{bio}}(t_b \mid \alpha_m, \omega_k) \right) \boldsymbol{d}$ denote the biological marginal likelihood of pattern p across branch b, and let $L_p = (1 - \epsilon_{\text{sink}}) L^{\text{bio}}_{b,p} + \epsilon_{\text{sink}} L^{\text{sink}}_{b,p}$ denote the total pattern likelihood. By Bayes' rule, the posterior responsibility that branch-site (b, p) was generated by the error sink is:

$$r^{\text{sink}}_{b,p} = \Pr(\text{sink} \mid b, p) = \frac{\epsilon_{\text{sink}} L^{\text{sink}}_{b,p}}{(1 - \epsilon_{\text{sink}}) L^{\text{bio}}_{b,p} + \epsilon_{\text{sink}} L^{\text{sink}}_{b,p}}. \tag{6}$$

In each Expectation-Maximization (EM) step, the maximum-likelihood update for ϵ_{sink} across all tested branches $\mathcal{B}_{\text{test}}$ ($|\mathcal{B}_{\text{test}}| = B_{\text{test}}$) and codon sites $C = \sum_{p=1}^{P} w_p$ is:

$$\epsilon^{\text{new}}_{\text{sink}} = \text{clip} \left(\frac{1}{B_{\text{test}} C} \sum_{b \in \mathcal{B}_{\text{test}}} \sum_{p=1}^{P} w_p r^{\text{sink}}_{b,p},\ 10^{-8},\ \epsilon_{\max} \right). \tag{7}$$

To prevent the biological positive-selection rate class ω_K from degenerating into an infinite-rate saturation kernel ($\omega_K = 2000$) on short branches and competing with $\boldsymbol{P}_{\text{sink}}$, HyPhy 3 caps the biological rate at $\omega_K \leq 50.0$ whenever `--error-sink` is active. Any instantaneous saturation jump on a short branch is therefore routed into $\boldsymbol{P}_{\text{sink}}$ under both the Alternative and Null models.

For each branch-site (b, s), HyPhy 3 reports the posterior error probability $r^{\text{sink}}_{b,s}$ and the Empirical Bayes Factor:

$$\text{BF}^{\text{sink}}_{b,s} = \frac{r^{\text{sink}}_{b,s} / (1 - r^{\text{sink}}_{b,s})}{\epsilon_{\text{sink}} / (1 - \epsilon_{\text{sink}})} = \frac{L^{\text{sink}}_{b,s}}{L^{\text{bio}}_{b,s}}. \tag{8}$$

Branch-sites with $\text{BF}^{\text{sink}}_{b,s} \geq 20.0$ and $r^{\text{sink}}_{b,s} \geq 0.50$ are reported in the console summary and JSON output alongside their inferred ancestral and descendant codons.

3 The `BUSTED --auto` 4-Gate Model Selection Pipeline

Instead of fitting all 36 combinations of (E, H, M, K), `BUSTED --auto` executes a single fast baseline fit ($K = 2, M = 1, \delta = 0, \psi = 0, \epsilon_{\text{sink}} = 0$) followed by a single Inside-Outside pass that caches the subtree vectors $D_{v,p}$ and ancestral complement vectors $V_{v,p}$ across all branches and patterns. It then evaluates four sequential diagnostic gates.

3.1 Gate 1: Error Sink Activation via Branch-Site Surprisal and Spatial Adjacency

At every tested branch b and site s (with pattern $p(s)$), Gate 1 computes the log-2 surprisal gain (in bits) of replacing the biological transition kernel with the saturation kernel $\boldsymbol{P}_{\text{sink}}$:

$$\Delta I^{\text{sink}}_{b,s} = \log_2 \left(\frac{L^{\text{sink}}_{b,p(s)}}{L^{\text{bio}}_{b,p(s)}} \right) \quad \text{bits}. \tag{9}$$

Under a well-specified single-hit biological model, $\Delta I^{\text{sink}}_{b,s} < 0$ at conserved sites and rarely exceeds 8 bits at rapidly evolving sites. A 2-nt or 3-nt misalignment on a branch of length $t_b = 0.002$ has $L^{\text{bio}}_{b,s} \propto t_b^2$ or t_b^3, yielding $\Delta I^{\text{sink}}_{b,s} \in [14, 35]$ bits. Gate 1 also counts spatially adjacent outlier pairs on the same branch:

$$N_{\text{adj}} = \sum_{b \in \mathcal{B}_{\text{test}}} \sum_{s=1}^{C-1} \mathbb{I} \left(\Delta I^{\text{sink}}_{b,s} > 6.0 \text{ and } \Delta I^{\text{sink}}_{b,s+1} > 6.0 \right), \tag{10}$$

which detects multi-codon indels and local frameshifts.

- **Screening rule:** If $\max_{b,s} \Delta I_{b,s}^{\text{sink}} < 10.0$ bits and $N_{\text{adj}} = 0$, the alignment has no saturation anomalies; `--error-sink` is turned **OFF** ($\epsilon_{\text{sink}} = 0$) in zero extra tree passes.
- **Confirmation probe:** Otherwise, Gate 1 runs a 3-step EM probe on $(\epsilon_{\text{sink}}, \boldsymbol{\omega}, \boldsymbol{p})$ starting from the baseline fit. It activates `--error-sink` if and only if the log-likelihood increases by $\Delta\ell_{\text{sink}} > 2.0$ (AIC_c improvement after penalizing 1 parameter).

3.2 Gate 2: Error-Quarantined Multi-Hit Score Test (δ, ψ)

To prevent alignment artifacts from triggering multi-nucleotide substitution parameters, Gate 2 weights each branch-site by its non-error probability $1 - r_{b,p}^{\text{sink}}$ from Gate 1. Let $\dot{\boldsymbol{Q}}_\delta = \partial \boldsymbol{Q}/\partial\delta|_{\delta=0}$ and $\dot{\boldsymbol{Q}}_\psi = \partial \boldsymbol{Q}/\partial\psi|_{\psi=0}$ denote the tangent generators for 2-nucleotide ($d(i,j) = 2$) and 3-nucleotide ($d(i,j) = 3$) transitions. Using the first-order short-branch tangent approximation $\partial \boldsymbol{P}_b/\partial\delta \approx t_b \dot{\boldsymbol{Q}}_\delta$, the error-quarantined positive tangent scores are:

$$S_\delta = \sum_{b \in \mathcal{B}_{\text{test}}} \sum_{p=1}^{P} w_p (1 - r_{b,p}^{\text{sink}}) \frac{t_b \, \boldsymbol{u}_{b,p}^T \dot{\boldsymbol{Q}}_\delta^+ \boldsymbol{d}_{b,p}}{L_{b,p}}, \tag{11}$$

$$S_\psi = \sum_{b \in \mathcal{B}_{\text{test}}} \sum_{p=1}^{P} w_p (1 - r_{b,p}^{\text{sink}}) \frac{t_b \, \boldsymbol{u}_{b,p}^T \dot{\boldsymbol{Q}}_\psi^+ \boldsymbol{d}_{b,p}}{L_{b,p}}, \tag{12}$$

where $\dot{\boldsymbol{Q}}_\delta^+$ and $\dot{\boldsymbol{Q}}_\psi^+$ contain the off-diagonal 2-hit and 3-hit rates.

- **Screening rule:** If $S_\delta < 1.5$, multi-hit parameters are set to `None` ($\delta = \psi = 0$) immediately.
- **Confirmation probe:** If $S_\delta \geq 1.5$, Gate 2 evaluates a 1D coordinate probe for $\delta \in \{0.05, 0.15, 0.30\}$ with ϵ_{sink} fixed from Gate 1. It enables `Double` only if $\Delta\ell_\delta > 2.0$ ($\Delta\text{AIC}_c > 2.0$), and subsequently enables `Double+Triple` only if $S_\psi \geq 1.0$ and $\Delta\ell_\psi > 2.0$.

3.3 Gate 3: Synonymous Rate Variation (M) via Neyman-Scott Overdispersion

To test whether a single synonymous rate ($M = 1$) is sufficient or whether site-to-site synonymous rate variation ($M \in \{2, 3\}$) is required, Gate 3 evaluates the empirical variance of site-level synonymous scaling gradients. At each pattern p, let $g_p = \partial \ln L_p/\partial\alpha_p|_{\alpha_p=1}$ denote the derivative of the site log-likelihood with respect to a site-specific synonymous rate multiplier α_p, evaluated analytically from the cached Inside-Outside vectors using $\partial \boldsymbol{P}_b/\partial\alpha_p = t_b \boldsymbol{Q} \boldsymbol{P}_b$. Under the homogeneous null $H_0 : \text{Var}(\alpha_s) = 0$, the Neyman-Scott overdispersion Z-statistic across C codons is:

$$Z_{\text{SRV}} = \frac{1}{\sqrt{2C}} \sum_{p=1}^{P} w_p \left[\frac{(g_p - \bar{g})^2}{\bar{v}} - 1 \right], \tag{13}$$

where $\bar{g} = C^{-1} \sum_p w_p g_p$ and $\bar{v}$ is the baseline squared gradient scale.

- **Screening rule:** If $Z_{\text{SRV}} < 1.5$, synonymous rate variation is turned **OFF** ($M = 1$).
- **Confirmation probe:** If $Z_{\text{SRV}} \geq 1.5$, Gate 3 runs a 15-iteration SQUAREM probe for $M = 2$ (2 extra parameters) and, if $Z_{\text{SRV}} > 3.5$, for $M = 3$ (4 extra parameters), selecting the $M \in \{1, 2, 3\}$ that minimizes AIC_c.

3.4 Gate 4: Non-Synonymous Rate Classes (K) via Constrained-Null Collapse Pruning

Using the selected (E, H, M) configuration, Gate 4 compares $K = 2$ ($\omega_1 \leq 1, \omega_2 \geq 1$) against $K = 3$ ($\omega_1 \leq \omega_2 \leq 1, \omega_3 \geq 1$) under the Constrained Null model ($\omega_K = 1.0$) before branch-length refinement.

Because selecting K under the Null model does not inspect the positive selection parameter $\omega_K > 1$, it preserves the asymptotic $\frac{1}{2}\chi^2_0 + \frac{1}{2}\chi^2_2$ distribution of the final Likelihood Ratio Test. Gate 4 selects $K = 3$ if and only if all three conditions hold:

1. $\mathrm{AIC}_c(\mathrm{Null}, K = 3) < \mathrm{AIC}_c(\mathrm{Null}, K = 2)$ ($\Delta\ell > 2.0$ for the 2 extra parameters);
2. Background rate separation: $|\hat{\omega}_2 - \hat{\omega}_1| \geq 0.05$;
3. Non-degenerate class weights: $\min_{k\in\{1,2,3\}} \hat{p}_k \geq 0.01$.

Otherwise, Gate 4 prunes the collapsed rate class and selects $K = 2$. Finally, HyPhy 3 performs full joint L-BFGS + SQUAREM optimization of the Unconstrained Alternative and Constrained Null models for the single winning configuration (E^*, H^*, M^*, K^*).

4 Verification and Empirical Benchmarks

Table 1 summarizes the behavior of Standard `BUSTED`, `BUSTED-E` (`--error-sink`), and `BUSTED --auto` across two validation benchmarks in `tests/test_busted.cpp`:

1. **Clean empirical alignment (*Drosophila Adh*, 23 taxa, 254 codons):** Gate 1 finds a maximum branch-site surprisal of 9.42 bits (< 10.0 bits) and leaves the Error Sink **OFF**. Gate 2 finds zero likelihood improvement from double hits ($\Delta\ell_\delta = 0$) and keeps Multi-Hits set to **None**. Gate 3 detects strong synonymous rate overdispersion ($Z_{\mathrm{SRV}} = 6.91$, $\Delta\ell_{\mathrm{SRV}} = +36.01$) and enables **SRV ($M = 3$)**. Gate 4 prunes the collapsed third background class and selects $K = 2$, detecting episodic positive selection with LRT = 25.70 ($p = 1.31 \times 10^{-6}$).
2. **Purifying alignment with two 3-nt misalignment artifacts (6 taxa, 100 codons, 2 corrupted codons on branch `T1` of length $t_b = 0.002$):** Standard `BUSTED` suffers a severe false positive ($\hat{\omega}_3 = 2000$, LRT = 16.65, $p = 1.21 \times 10^{-4}$). Under `BUSTED --auto`, Gate 1 detects a 19.19-bit surprisal spike ($\Delta\ell_{\mathrm{sink}} = +5.55$) and turns the Error Sink **ON** ($\hat{\epsilon}_{\mathrm{sink}} = 0.00226$ vs. true $2/900 = 0.00222$). Because Gate 2 weights the multi-hit score by $(1 - r^{\mathrm{sink}}_{b,s})$, the two 3-nt errors are quarantined ($S_\delta = 0.058 < 1.5$), keeping Multi-Hits **None**. Both corrupted branch-sites (`T1` codon 10, BF = 1.06×10^6; `T1` codon 50, BF = 1.38×10^5) are flagged, and the false positive is eliminated (LRT = 0.0, $p = 0.50$).

5 Command-Line and JSON Interface

Both `BUSTED --auto` and `--error-sink` are available in the HyPhy 3 command-line interface.

- **Full automatic 4-gate model selection (recommended default):** `hyphy3 busted --alignment gene.nex --auto --output gene.BUSTED.json`.
- **Explicit saturation error sink with custom Bayes factor threshold:** `hyphy3 busted --alignment gene.nex --error-sink --error-sink-bf 20.0`.

The modern JSON output records the 4-gate diagnostic metrics under `"auto_model_selection"` and any flagged branch-site error events under `"error_sink_diagnostics"`.

Table 1: Validation of `BUSTED`, `BUSTED-E` (`--error-sink`), and `BUSTED --auto` across clean and error-contaminated alignments.

Dataset	Method	Sink (E)	MH (H)	SRV (M)	K	LRT (p-value)	Diagnostic Outcome
Drosophila Adh	Standard `BUSTED`	Off	None	1	3	33.75 (2.3×10^{-8})	Baseline fit ($\ln L = -4703.29$).
(23 taxa, 254 cod.)	`BUSTED-S`	Off	None	3	3	16.69 (1.2×10^{-4})	Manual SRV ($\ln L = -4687.58$).
	`BUSTED --auto`	**Off**	**None**	**3**	**2**	**25.70** (1.3×10^{-6})	Gate 3 enables SRV ($Z = 6.91$, $\Delta\ell = +36.0$); Gate 4 prunes collapsed $K = 3 \to 2$.
Purifying + 2 Errors	Standard `BUSTED`	Off	None	1	3	16.65 (1.2×10^{-4})	**False positive** ($\hat{\omega}_3 = 2000$ forced by two 3-nt errors on $t_{\mathrm{T1}} = 0.002$).
(6 taxa, 100 cod.)	`BUSTED-E`	On	None	1	3	0.10 (0.476)	Quarantines codons 10 & 50 on `T1` ($\hat{\epsilon}_{\mathrm{sink}} = 0.00226$).
	`BUSTED --auto`	**On**	**None**	**1**	**2**	**0.00** (0.500)	Gate 1 activates Sink (19.2 bits, $\Delta\ell = +5.55$); Gate 2 quarantines MH ($S_\delta = 0.058$).